Starting – General Data

1. General

Engine starting is achieved by a starter/generator in conjunction with a START switch, control relays and the ignition system (refer to 74-00-00, Ignition – General Data). Electrical power for starting is provided by the aircraft battery (main) or, if available, an external power source. DC power from the aircraft battery or external power source, through the reverse current relay, drives the starter/generator which rotates the gas generator turbine up to approximately 12% rpm, when the fuel control lever is operated to permit light up to occur. When the engine has accelerated to a self-sustaining speed, and the START switch is released, the starter/generator may be selected "on-line" for DC generation (refer to 24-30-00, DC Generation – General Data). An automatic fuel purge valve (refer to 71-70-00, Engine Drains – General Data) is provided for each engine. When the relevant engine start circuit is energized, the valve is energized to provide a means of purging trapped air from the appropriate engine fuel control unit; the valve is de-energized when the START switch is released. For engine starting procedures refer to 71-00-00, Power Plant (Mod 6/2012) – General Data.

Aircraft with Mod 6/2315 also include a small 24 volt avionics start battery. The battery is used to supplement electrical power supply to certain avionics systems during an engine start. Further information on the avionics start battery is provided in 24-30-32, Avionics Start Battery (Mod 6/2315) – General Data.

2. Description

The starter system consists of a starter/generator mounted on the accessory case of each engine, a START switch marked LEFT–RIGHT located on the overhead console in the flight compartment and control relays (left and right start relays, and left and right No. 1/No. 2 control relays) located in the main distribution box. Circuit protection is provided by two 5 ampere circuit breakers, labelled START L and START R, on the main circuit breaker panel.

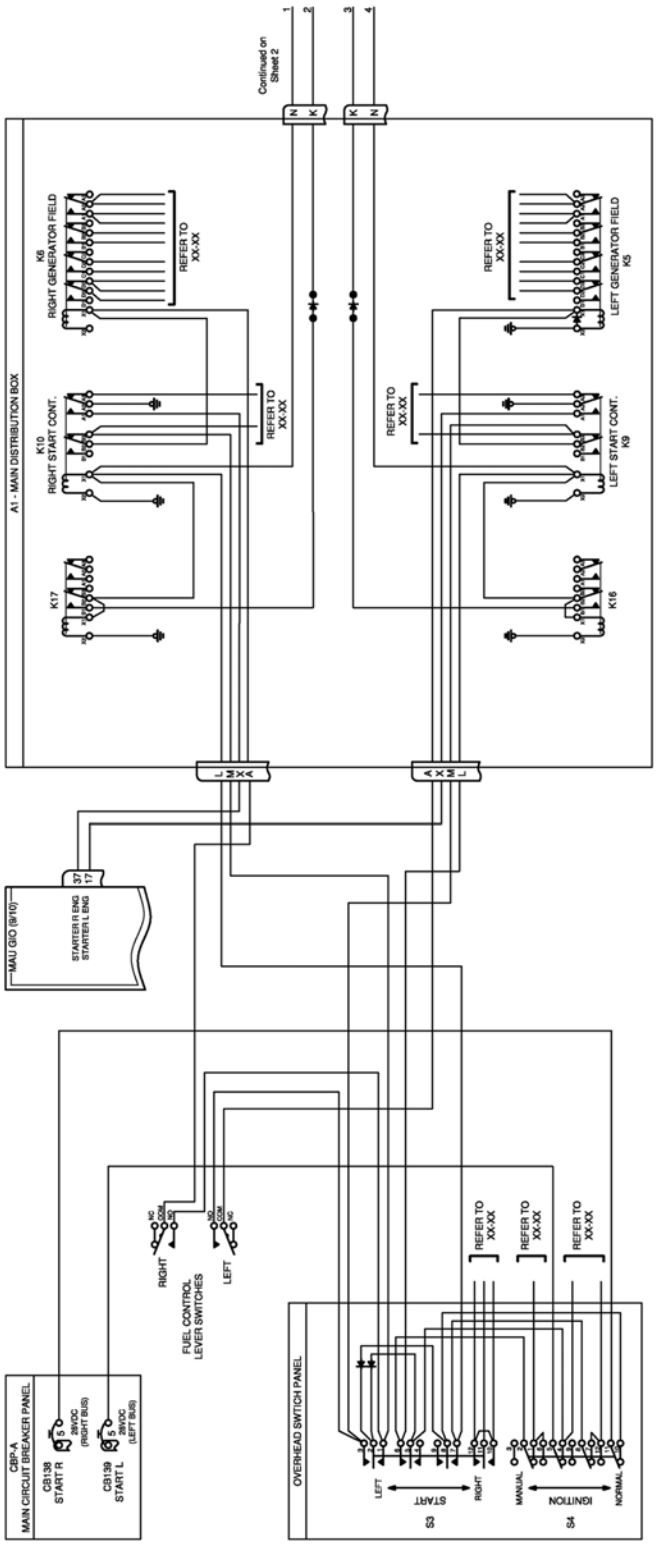
3. Operation

Refer to Figure 1 for details.

With the MASTER switch set to ON, and the POWER SOURCE switch selected to BATTERY or EXTERNAL (Pre Mod 6/2314) or with EXTERNAL or BATTERY switches ON (Post Mod 6/2314), depending on power supply available, the main battery relays (or external power relays) are energized, connecting battery or external power to the DC buses.

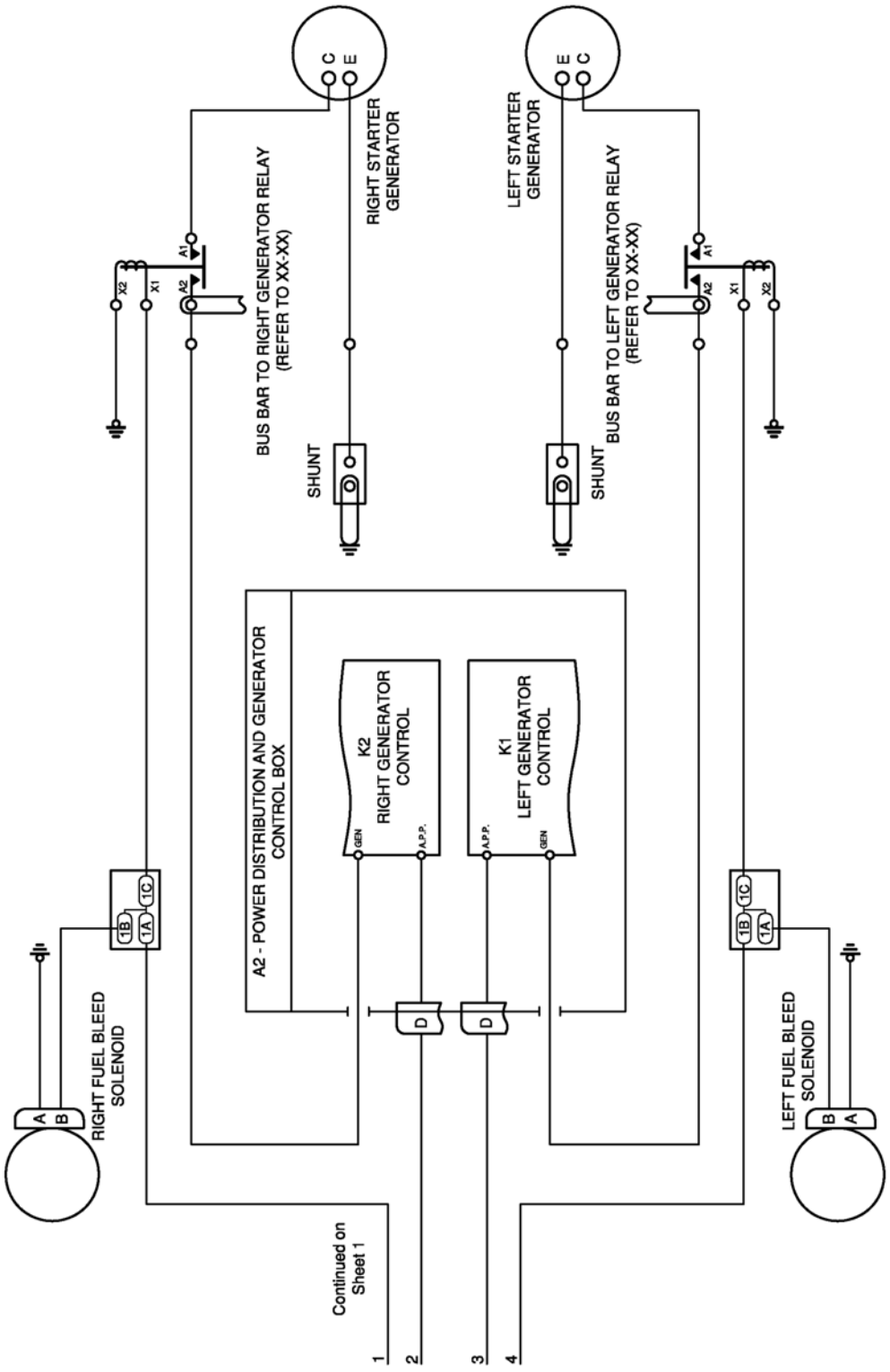
With the IGNITION switch set to NORMAL, and the START switch held to LEFT or RIGHT, DC power from the battery bus is fed through the IGNITION and START switches to energize the start relay, generator field relay (through the fuel control lever limit switch), ignition and start relay, start No. 1 and No. 2 control relays and the fuel purge valve solenoid. With the start No. 2 control relay energized, power is connected to the APP terminal of the reverse current relay, energizing the main relay to connect main battery (or external) power to the starter/generator (through the reverse current relay main contacts and the closed contacts of the start relay), which operates to rotate the gas generator turbine. (The START switch overrides the generator field relays to hold the external power relay on).

When the gas generator turbine rpm stabilizes at approximately 12% and, in accordance with the starting procedure (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data) the fuel control lever is advanced to ON to provide fuel to permit light-up, the fuel control lever limit switch opens. Opening the limit switch de-energizes the generator field relay which changes the starter/generator from a compound to a series motor for faster acceleration.



Engine Starting – Electrical Schematic
Figure 1 (Sheet 1 of 2)

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Engine Starting – Electrical Schematic
Figure 1 (Sheet 2 of 2)

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